

The IoT applications in India across industry sectors can be bucketed under three broad categories: consumer, industrial and public sector.

Consumer applications of the IoT. While organisations expect the IoT to increase automation and therefore improve productivity, consumers expect it to improve the overall quality of life. A research conducted by Tata Communications, presented in Fig. 3, showcases consumer expectations from the IoT.

The perception of the common people regarding the IoT is limited. A large number (~ 35 per cent) of people from the same survey mentioned above relate the IoT with only smartphones. Although this is not incorrect, the IoT is much more than just phones. Only ~ 14 per cent of all people surveyed (Fig. 4) mentioned all connected devices.

Although most of the current developments in the IoT are focussed on industrial opportunities, there has been traction in the consumer space, too. The IoT applications for consumers are opening new and intimate entertainment experiences. Shifting consumer preferences towards smart sensors, wearables and clustered systems, like fitness trackers or smart homes, are triggering growth in the consumer application of the IoT.

Smartphone applications that can lock doors, reset thermostats and more already exist. Firms are coming up with ways of cooking that can be done and monitored from the comfort of the living room sofa through the smartphone. Increasing sophistication of sensors makes it possible for devices to understand consumers at unprecedented levels.

Industrial applications of the IoT. Globally as well as in India, industrial adoption of the IoT has far exceeded consumer applications. The term Industry 4.0 heralds the coming of a new industrial revolution through smart manufacturing. The IoT is expected to drive operational efficiencies through automation, connectivity and analytics.

Industrial applications broadly include improved automation, efficient tracking and effective management through connected systems. In India, such sectors as manufacturing, automotive, transportation and logistics are experiencing the highest adoption

TABLE II
POTENTIAL CONSUMER-FOCUSSED APPLICATIONS OF THE IoT

Application	Brief description
Personalisation/customisation	• On-demand access to tailored content
Authentication/verification	• IoT sensors for multi-level authentication removing the need for painful password-based authentication
Wearables for better measurement	• Wearables have the potential to unlock new data that can both address deficiencies in the current measurement system • Through the IoT, advertisers and media companies may be able to answer critical questions about consumer behaviour

levels of the IoT. Typical industrial applications include:

- Manufacturing/supply chain: Improved process automation, connected factory, robotics, tracking of goods, tools, etc
- Agriculture: Handheld devices to figure out moisture/pH factor of soil, etc
- Transportation: Improved vehicle tracking, traffic management, etc
- Energy: Efficient management of energy usage, potential faults, etc

Public sector applications of the IoT. The IoT has significant applications in the public sector as well. The chart given in Table III lays down the potential end use of the IoT in this sector. In India, the government's push for smart cities is expected to be

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According to an IDC report, more than 50 per cent of India's corporate leaders believe the IoT to be imperative to remaining digitally fit, whereas 60 per cent see it as being imperative to driving a competitive advantage.

a key enabler for IoT adoption across a large number of these areas.

Challenges to be addressed

Even though the IoT is creating, and will likely continue to create, immense opportunities for both industry and consumers, it is not without its share of risks.

As is common when anything new comes onto the stage, legislators across the world are looking for ways to incorporate new regulatory frameworks to plug relevant holes of the IoT and ensure its smooth functioning. To realise the full potential of the IoT, businesses and governments need to work together to overcome the impediments that lay in its path.

Cyber security has already gained notoriety across the world due to increased incidents of cyber-attacks, espionage and data breaches. With the number of connected devices increasing and virtual and physical worlds seamlessly integrating, present security measures will no longer remain adequate. Organisations will require enhanced security frameworks from device-level authentication and application security to system-wide assurance, resiliency and incidence response models.

Another critical hurdle is the lack of interoperability among existing systems, which will significantly increase the complexity

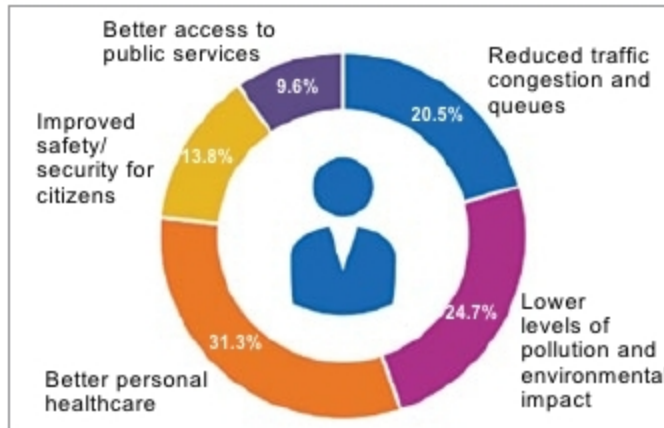


Fig. 3. Indian consumer expectation from the IoT
(Source: Tata Communications, 2018)

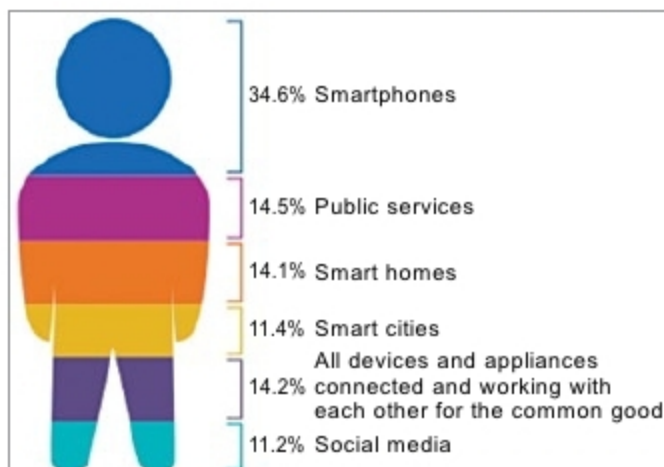


Fig. 4: Consumer insight by the IoT association
(Source: Tata Communications, 2018)